

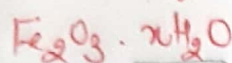
1 Mark

Define Corrosion

It is the process of gradual decay or deterioration of a metal from metal surface due to unwanted chemical or electrochemical interaction of metal with environment.

eg.

Rusting of Iron



! Mark
special kind of corrosion in case of Fe.
(formation of a layer of reddish scale of hydrated Fe_2O_3 on the surface of iron).

✓ Tarnishing of Silver (Ag) :- Black layer formation of Ag_2O

Effects of Corrosion

- (i) Loss of Product
- (ii) Loss of efficiency, due to loss of useful properties of metals.
- (iii) Will lead to leakage of Toxic liq. or gas, which may cause Fire Hazards.
- (iv) Increase in maintenance and production cost.
- (v) Contamination of product.

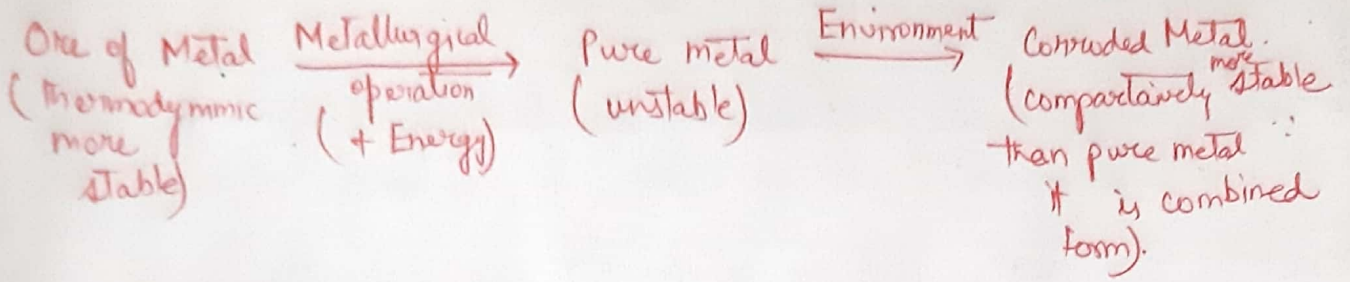
Cause of Corrosion

Metal exist in nature in the form of oxides, sulphides, sulphates, carbonates. (These combined state of metal are known as ores.)

-An ore is thermodynamical more stable b/c of low energy.
So metallurgy is a process in which metal is extracted from ore by providing considerable amt of energy so the extracted metal has high energy $\therefore$ thermodynamically unstable.

So it is a natural tendency of metal to go back to more stable state which is achieved in the form of corrosion. (intⁿ b/w metal and environment).

∴ corrosion is reverse of the metal extraction process.



Corrosion Losses

India loses approx. one tonne of steel every 15 sec. on account of corrosion.

(Losing 12 trillion (10^{12}) every year.)

Imp

due to corrosion in various sectors including infrastructure, utility service, manufacturing, defence and nuclear sectors.

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Units ∴ ∵ corrosion is a measure of thickness of oxide layer formed so it has same units of length.

Types and mechanism of Corrosion

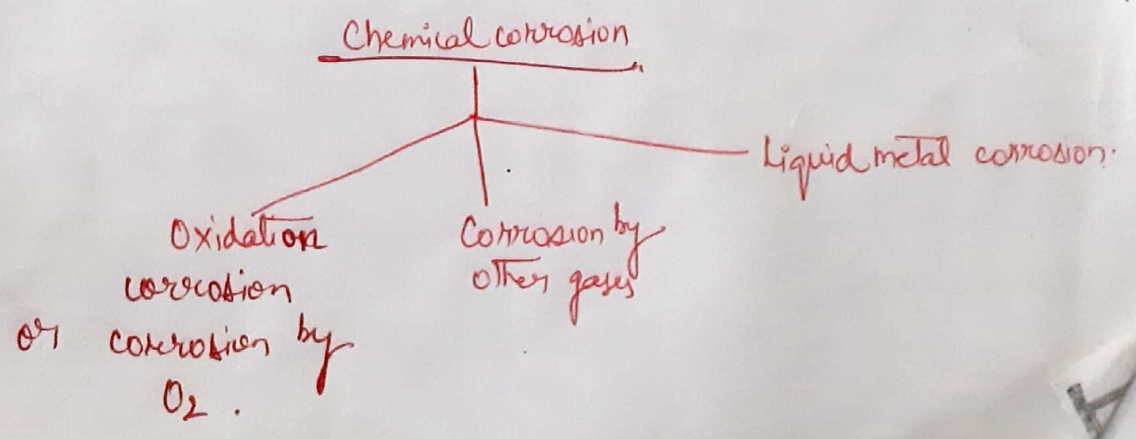
Two types

wet or electrochemical

Chemical or dry corrosion

- simplest case of corrosion
- Takes place by direct chemical attack.

When metal surface is in immediate proximity to atmospheric gases or anhydrous liquid, then chemical or dry corrosion occurs
(i.e. O_2, Cl_2, H_2S, N_2)



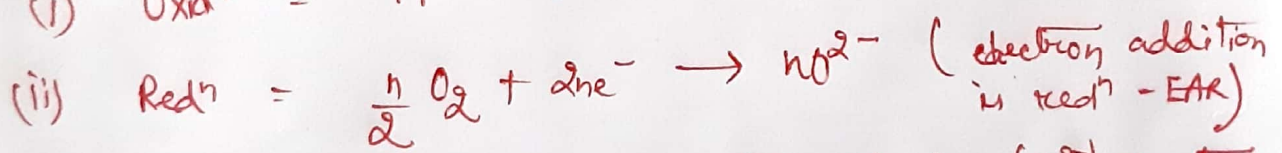
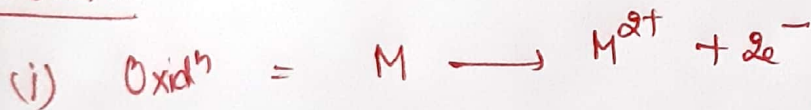
- example (i) Ag materials undergoes chemical corrosion by ⁽³⁾ atmosphere H_2S gas
- (ii) Fe undergoes chemical corrosion by HCl gas.

(i) Oxidation Corrosion (by O_2) :- It is done by the direct attack of O_2 at low or high temp. on metal surfaces in the absence of moisture.

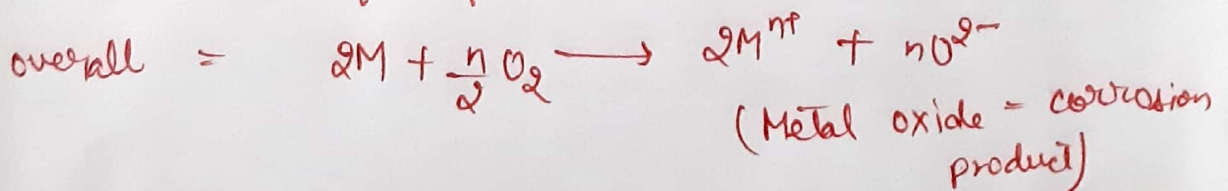
Alkali Metals (Li, Na, K etc) } are rapidly oxidised at
Alkaline earth metals (Mg, Ca, Sn, etc) } Low temp.

Almost all metals (except Ag, Au and Pt) } are oxidised at high temp.

Mechanism:-



Oxygen is converted to oxide ion (O^{2-}) due to transfer of e^- 's from metal.



When $oxid^n$ starts, a thin layer of oxide is formed on the metal surface and the nature of this film decides the further action.

Pilling Bedworth Rule :- According to it "any metal oxide is protective or non-porous if the volume of the metal oxide is atleast in $excess$.

as great as

upon the ~~ratio~~ of volume of the metal oxide to the volume of the metal consumed. (If it is > 1 it is protective)

This is known as Pilling-Bedworth ratio.

$$\frac{\text{Volume of metal oxide formed}}{\text{Volume of the metal consumed}}$$

If the ratio < 1 , then the layer of oxide will be Non-protective
e.g. alkali and alkaline earth metals such as Na, Ca, Mg (except Al).

Less than 1 means oxide film volume is less as compared to the metal from which it is formed, $\therefore$ cracks developed in oxide film and O_2 diffuse inwards further to attack metal for further corrosion.

If the metal oxide film formed is equal or greater than unity.

i.e. If the ratio > 1 , then the layer will be protective in nature.
e.g. Al, Cr, W etc.

so the metal oxide layer will be non-porous and O_2 will not diffuse inwards and rate of corrosion $\downarrow$.

Corrosion by other gases:-

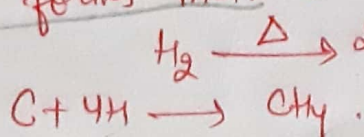
In addition to O_2 ; CO_2 , SO_2 , Cl_2 , H_2S , F_2 etc. also have corrosive effect on metals. The extent of corrosion depends on the chemical affinity b/w the metals and the gas involved.

2 Marks

Decarboxisation Process

The $\downarrow$ in the % of C in steel is called decarboxisation process.

When steel is exposed to Hydrogen environment at high temp. atomic hydrogen is formed which reacts with C, present in steel and forms methane.



This collection of methane in the voids of steel and $\downarrow$ in the % of C results in the cracks in steel.

Wet or electrochemical corrosion

When corrosion takes place by the transfer of e^- 's from anodic parts of metal to cathodic part through a conductive solution is called wet / electrochemical corrosion.

• It is more common than dry corrosion.

• It takes place mostly under wet or moist conditions.

Mechanism:

1. Separate anode and cathode b/w which current flows through the conducting medium.
2. Occurrence of oxidⁿ (corrosion) at anode, it generates metallic ions.
3. Non-metallic ions like OH^- or O^{2-} are formed at cathode.
4. Diffusion of metallic and non-metallic ions towards each other through conducting medium and formation of corrosion product somewhere b/w anodic and cathodic areas. (near the cathode cations are smaller so diffuse rapidly to large anions).

for example Rust of iron in acidic and neutral medium.

In acidic medium. (evolution of H_2)

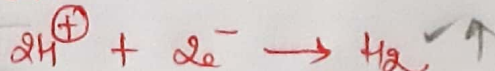
or alkaline medium

anode $\div$ $Fe \rightarrow Fe^{2+} + 2e^-$
(oxidⁿ) Iron by losing e^- 's and become ferrous.

cathode $\div$ Released e^- 's are conducted to cathode from anode.

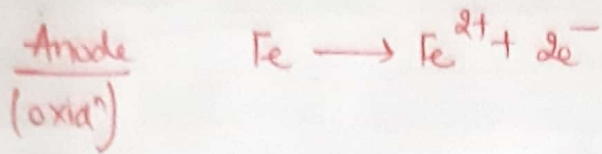
(redⁿ)

(i) When O_2 is absent



(ii) When O_2 is dissolved in acidic med. $4H^+ + O_2 + 4e^- \rightarrow 2H_2O$

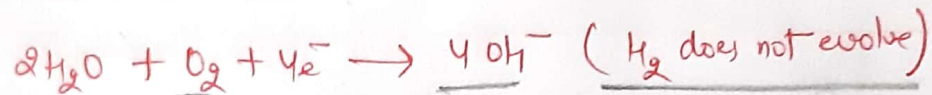
In neutral medium / alkaline medium



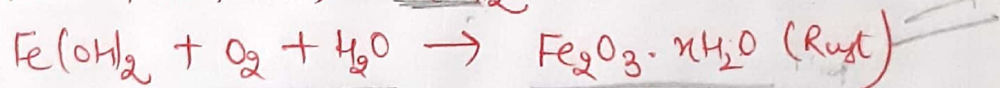
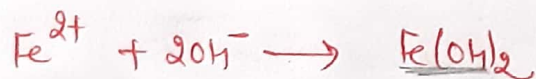
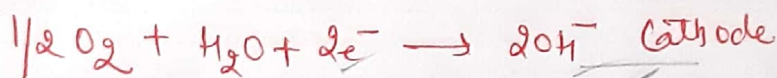
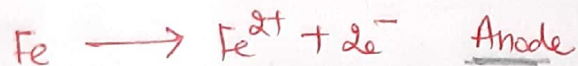
Cathode In absence of O_2



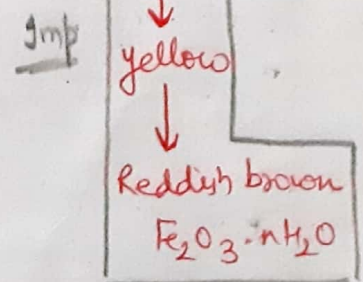
In presence of O_2



Example :- Rusting of iron in neutral aq. solⁿ in presence of O_2



Imp degree of hydration affects the color of the rust which may vary from Black Fe_2O_4



Note :- corrosion occurs at anode but rust is formed near the cathode b/c smaller ions Fe^{+2} produced at anode diffuse rapidly towards the cathode as compared to diffusion of larger OH^{-} ions towards anode.

Ques How does anode and cathode form takes place at metal surface?

Ans (i) Non-uniform or rough surface of metal (ii) presence of impurities in the bulk of metal (iii) Difference in O_2 con^t along the metal surface.

Types of Corrosion

1. Galvanic Corrosion or bimetallic corrosion or differential Metal Corrosion
2. Concentration cell corrosion
3. Water line corrosion
4. Pitting Corrosion
5. Crevices corrosion
6. Stress corrosion
7. Soil or Microbial corrosion

① Galvanic or bimetallic corrosion

This occurs when two dissimilar metals come in contact with other and corrosive atmosphere. (basically exposed to an electrolyte).
These two metallic parts constitute a "galvanic cell".

The more active metal out of two gets corroded.

oxidⁿ
redⁿ
One form anode (Low redⁿ potential)
One form cathode (High redⁿ potential)
-ve the value of redⁿ potential, more easily it can loose e⁻s. and gets oxidised.

example :- Zn and Cu are in contact with each other in presence of an electrolyte.

$E^\circ = -0.76V$ Zn (higher in electrochemical series with lower electrochemical potential) = Anode $Zn \rightarrow Zn^{2+} + 2e^-$
 $\therefore$ Zn gets corroded (oxidⁿ occurs)
 $E^\circ = +0.34V$ Cu (lower in electrochemical series with high electrochemical potential) = Cathode $Cu^{2+} + 2e^- \rightarrow Cu$
(Less noble is attacked)
Zn corrode less in HCl but in contact with Cu, corrodes more.

Larger the electrochemical potential b/w the two metals, greater will be the galvanic corrosion.

example :- Use of Cu pipes in conjugation with Fe pipes, Fe gets corroded and Cu is protected.

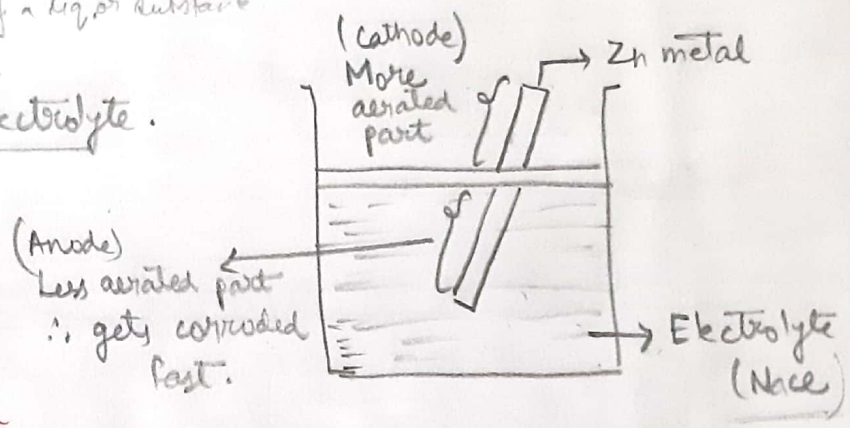
Prevention of galvanic corrosion

- (i) By avoiding galvanic couple
- (ii) By providing an insulating material b/w two metals.

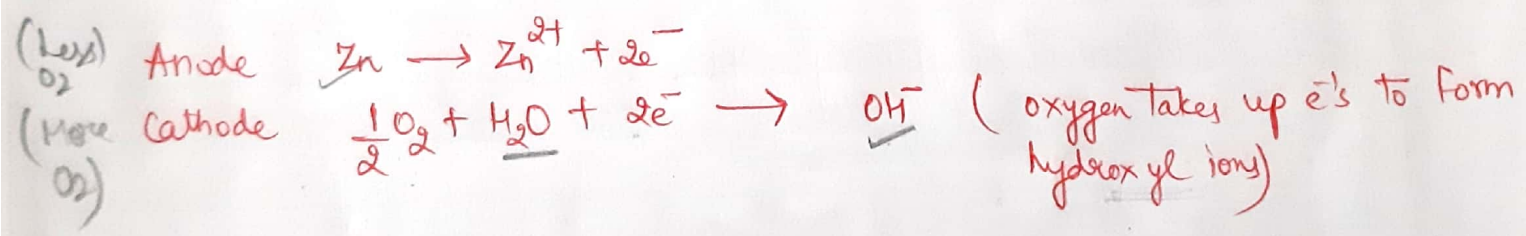
② Concentration Cell corrosion or differential aeration corrosion.

This type of corrosion takes place when two or more areas of a metal surface are in contact with different con^n of the same electrolyte or they are exposed to an electrolyte of varying aeration.

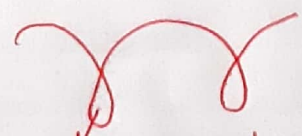
Different aeration means metal is partly immersed in an electrolyte.
 $\rightarrow$ Presence of O_2 in a liq. or substance



Due to diff. con^n of O_2 , potential is set up which causes a flow of current b/w two areas of same metal



example:- wire fence


These parts are less accessible to air than rest of the part $\therefore$ acts as anode and undergo corrosion.

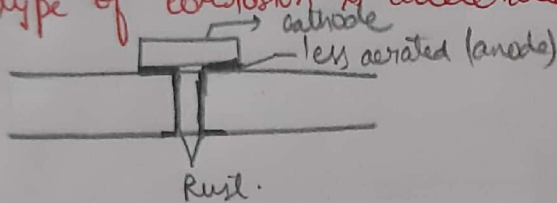
(Junction) (3) Service Corrosion :-

- Also known as contact corrosion.
- It is the corrosion produced at the region of contact of metals with metals or non-metals.

example it occurs at the junction of two metals such as bolts, nuts, washers etc.

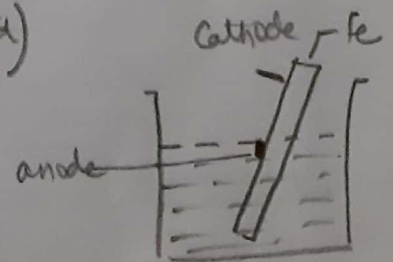
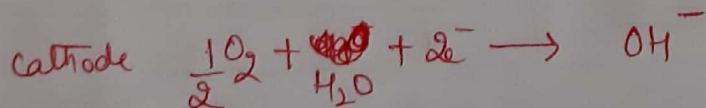
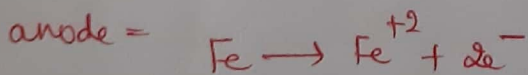
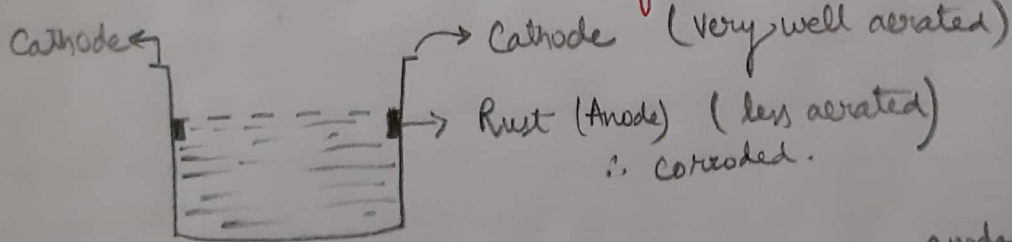
These junctions becomes = anode b/c of less O₂ supply (corrosion occurs)
(∵ less exposed area)
Exposed = cathode area

This type of corrosion is accelerated by the deposition of dirt etc.



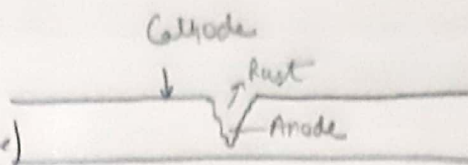
(4) Water line corrosion :- Corrosion along a line just beneath the level of H₂O stored in an iron tank.

This is b/c of differential aeration.



5. Pitting Corrosion

(limited to a particular place)



- It is localized corrosion
- Results in the formation of pits or holes.
- This is due to cracking of the protective film on the metal surface.
- Broken layer = Anode = $\text{Fe} \rightarrow \text{Fe}^{2+} + 2e^-$
 Unbroken " = Cathode $\frac{1}{2}\text{O}_2 + 2\text{H}_2\text{O} + 2e^- \rightarrow 2\text{OH}^-$

Formation of small anode and large cathodic areas result in severe corrosion of the anode region.

Rust further screens the bottom of the pit from O_2 $\therefore$ Ting The rate of corrosion.

6. Stress corrosion (Stress cracking)

Decay / destruction of material due to combined activity of

- mechanical stress (different types of uneven stresses such as rolling, drawing or insufficient annealing)
- corrosive environment (even traces of corrosive environment)

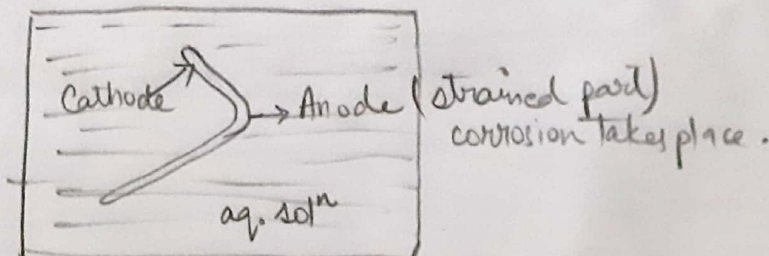
stress area = anode ✓

another area = cathode.

used in
Hardware

examples fabricated articles of some alloys such as Zn brasses or Ni brasses

Annealing :- Heat the metal and allow to cool slowly.



⑦ Microbial or soil corrosion

Decay / Deterioration of materials caused directly or indirectly by microbes such as bacteria, algae, moulds or fungi.

Buried pipes which are extended from one type of soil to another have this type of corrosion due to

- (i) differential aeration
- (ii) Microbes present.

<u>example</u>		<u>sand</u>	or	<u>clay</u>
		↓		↓
		More H_2O_2		Less O_2
		Cathode.		Anode

✓ Passivity (this is due to the formation of highly protective and thin film on metal surface).
(resistant to corrosion)

This refers to the loss of chemical reactivity of certain metals and alloys under particular condition.

- They become inert and act as noble metals
- So there is strong reduction of corrosion due to formation of protective layer example Al_2O_3 , Cr, Sn, Ni, Ti etc.

Significance . This has been used successfully to develop new methods for prevention of corrosion.

- And to predict corrosion resistance.

Galvanic Series

In electrochemical series, The metal at the top are more anodic, $\therefore$ undergoes corrosion faster than the metal which is at bottom of electrochemical series.

However due to passivity (formation of a protective layer of oxide) sometimes these metals behave exactly opposite to that predicted by the electrochemical series.

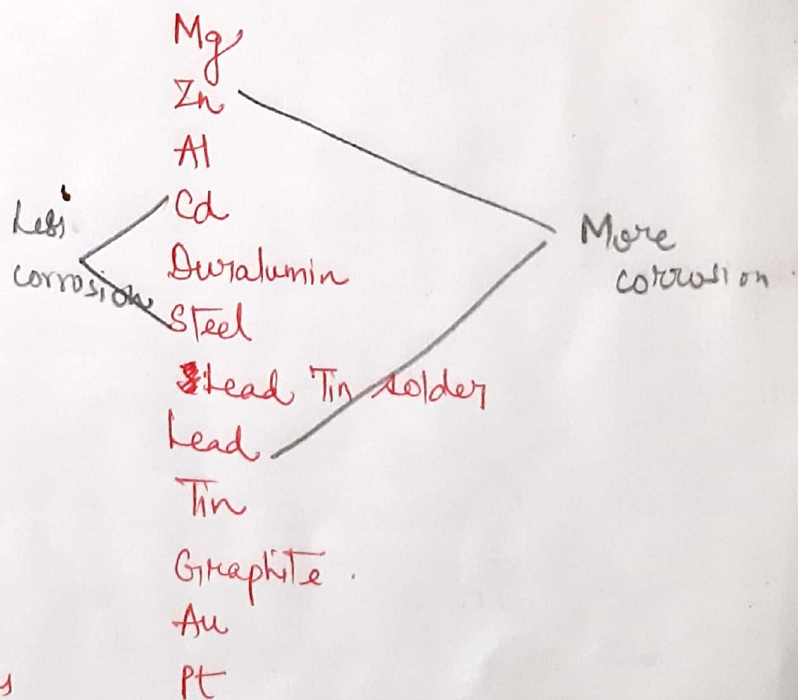
So, for predicting gc corrosion, galvanic series is required

Metal having -ve electrode potential = Anode ✓

Metal having +ve electrode potential = Cathode ✓

Anodic / Most Active

greater is the diff. b/w two metals more is corrosion.



Cathode / Noble Metals

FACTORS INFLUENCING CORROSION

There are two factors that influence the rate of corrosion. Hence a knowledge of these factors and the mechanism with which they affect the corrosion rate is essential because the rate of corrosion is different in different atmosphere.

1. Nature of the metal

2. Nature of the corroding environment

Nature of the metal:

a) **Physical state:** The rate of corrosion is influenced by physical state of the metal (such as grain size, orientation of crystals, stress, etc). The smaller the grain size of the metal or alloy, the greater will be its solubility and hence greater will be its corrosion. Moreover, areas under stress, even in a pure metal, tend to be anodic and corrosion takes place at these areas.

b) **Purity of metal:** Impurities in a metal cause ^{non-uniform} heterogeneity and form minute/tiny electrochemical cells (at the exposed parts), and the anodic parts get corroded. The cent percent pure metal will not undergo any type of corrosion. For example, the rate of corrosion of aluminium in hydrochloric acid with increase in the percentage impurity is noted.

% purity of aluminium	99.99	99.97	99.2
Relative rate of corrosion	1	1000	30000

Billal Billal

d) **Nature of surface film:** In aerated atmosphere, practically all metals get covered with a thin surface film (thickness = a few angstroms) of metal oxide. The ratio of the volumes of the metal oxide to the metal is known as a specific volume ratio. Greater the specific volume ratio, lesser is the oxidation corrosion rate. The specific volume ratios of Ni, Cr and W are 1.6, 2.0 and 3.6 respectively. Consequently the rate of oxidation of tungsten is least, even at elevated temperatures..

e) **Relative areas of the anodic and cathodic parts:** When two dissimilar metals or alloys are in contact, the corrosion of the anodic part is directly proportional to the ratio of areas of the cathodic part and the anodic part.

Corrosion is more rapid and severe, and highly localized, if the anodic area is small (eg., a small steel pipe fitted in a large copper tank), because the current density at a smaller anodic area is much greater and the demand for electrons can be met by smaller anodic areas only by undergoing corrosion more briskly.

f) **Position in galvanic series:**

g) **Passive character of metal:**

h) **Solubility of corrosion products:**

i) **Volatility of corrosion products:**

*↓ area more less corrosion
Anode, more less corrosion
Cathode are more corrosion*

Nature of the Corroding Environment:

a) **Temperature:** The rate of corrosion is directly proportional to temperature i.e., rise in temperature increases the rate of corrosion. This is because the rate of diffusion of ions increases with rise in temperature.

b) **Humidity of air:** The rate of corrosion will be more when the relative humidity of the environment is high. The moisture acts as a solvent for oxygen, carbon dioxide, sulphur dioxide etc. in the air to produce the electrolyte which is required for setting up a corrosion cell.

c) **Presence of impurities in atmosphere:** Atmosphere in industrial areas contains corrosive gases like CO_2 , H_2S , SO_2 and fumes of HCl , H_2SO_4 etc. In presence of these gases, the acidity of the liquid adjacent to the metal surfaces increases and its electrical conductivity also increases, thereby the rate of corrosion increases.

d) **Presence of suspended particles in atmosphere:** In case of atmospheric corrosion: (i) if the suspended particles are chemically active in nature (like NaCl , Ammonium sulphate), they absorb moisture and act as strong electrolytes, thereby causing enhanced corrosion; (ii) if the suspended particles are chemically inactive in nature (eg., charcoal), they absorb both sulphur gases and moisture and slowly enhance corrosion rate.

e) **Influence of pH:** Generally acidic media (ie., $\text{pH} < 7$) are more corrosive than alkaline and neutral media. However, amphoteric metals (like Al , Zn , Pb , etc.) dissolve in alkaline solutions as complex ions. The corrosion rate of iron in oxygen-free water is slow, until the pH is below 5. The corresponding corrosion rate in presence of oxygen is much higher. Consequently corrosion of metals, readily attacked by acid, can be reduced by

increasing the pH of the attacking environment, eg., Zn (which is rapidly corroded, even in weakly acidic solutions such as carbonic acid suffers minimum corrosion at pH=11.

- f) Nature of ions present:
- g) Conductance of the corroding medium:
- h) Formation of oxygen concentration cell:
- i) Flow velocity of process stream:
- j) Polarization of electrodes:

CORROSION CONTROL (PROTECTION AGAINST CORROSION)

As the corrosion process is very harmful and losses incurred are tremendous, it becomes necessary to minimize or control corrosion of metals. Corrosion can be stopped completely only under ideal conditions. But the attainment of ideal conditions is not possible. However, it is possible only to minimize corrosion considerably. Since the types of corrosion are so numerous and the conditions under which corrosion occurs are so different, diverse methods are used to control corrosion. As the corrosion is a reaction between the metal or alloy and the environment, any method of corrosion control must be aimed at either modifying the metal or the environment.

a. Choice of metals and alloys:

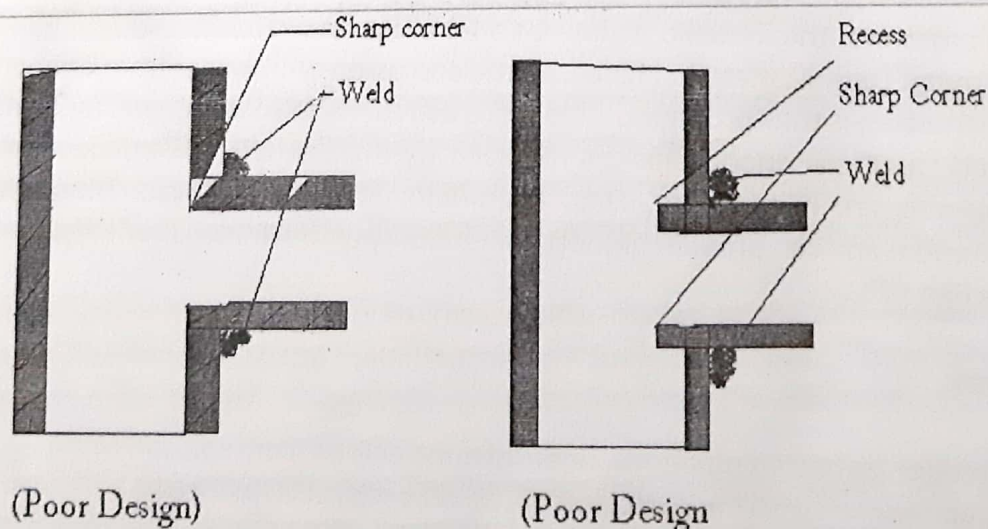
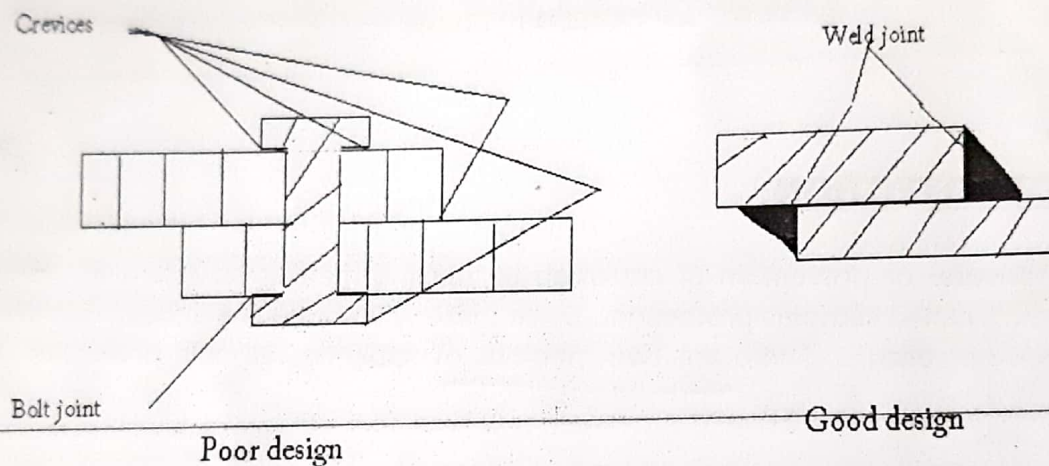
1. The first choice is to use noble metals such as gold and platinum. They are most resistant to corrosion. As they are precious, they cannot be used for general purposes.
2. The next choice is to use purest possible metal. But in many cases, it is not possible to produce a metal of high chemical purity. Hence, even a trace amount of impurity leads to corrosion.
3. Thus, the next choice is the use of corrosion resistant alloys. Several corrosion resistant alloys have been developed for specific purposes and environment. For example, a) Stainless steel containing chromium produce an exceptionally coherent oxide film which protects the steel from further attack. (b) Cupro-nickel (70% Cu + 30%Ni) alloys are now used for condenser tubes and for bubble trays used in fractionating column in oil refineries. (c) Highly stressed Nimonic alloys (Ni-Cr-Mo alloys) used in gas turbines are very resistant to hot gases.

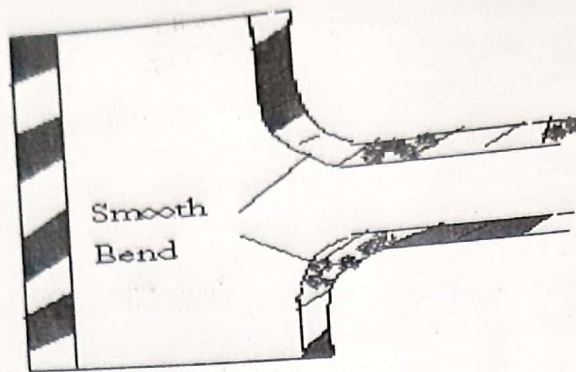
b. Proper Designing:

Proper geometrical design plays a vital role in the control of corrosion of equipments and structures. The general guidelines of the design of materials and components to control corrosion are the following:

- a. Use always simple design and structure
- b. The design must avoid more complicated shapes having more angles, edges, corners etc.
- c. Avoid the contact of dissimilar metals as they may lead to galvanic type corrosion. To overcome this, insulation can be used.

- d. When two dissimilar metals are to be in contact, the anodic area must be as large as possible and the cathodic area should be as small as possible.
- e. As far as possible, crevices (gap or crack) should be avoided between adjacent parts of a structure.
- f. Bolts and rivets should be replaced by proper welding
- g. Metal washers should be replaced by rubber or plastic washers as they do not adsorb water. They also act as insulation.
- h. Corrosion in pipelines can be prevented by using smooth bends.
- i. Heat treatment like annealing minimizes the stress corrosion.
- j. A good design of water storage container is the one from which water can be drained and cleaned easily. Such a design avoids accumulation of dirt etc.





(Best Design)

CATHODIC PROTECTION:

The reduction or prevention of corrosion by making metallic structure as cathode in the electrolytic cell is called cathodic protection. Since there will not be any anodic area on the metal, corrosion does not occur. There are two methods of applying cathodic protection to metallic structures.

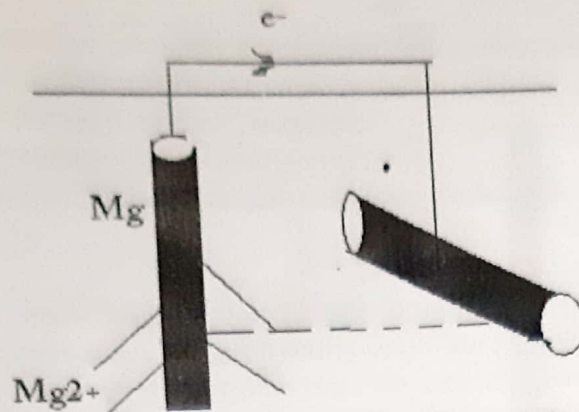
- a) Sacrificial anodic protection (galvanic protection)
- b) Impressed current cathodic protection

a) SACRIFICIAL ANODIC PROTECTION METHOD

In this method, the metallic structure to be protected is made cathode by connection it with more active metal (anodic metal). Hence, all the corrosion will concentrate only on the active metal. The parent structure is thus protected. The more active metal so employed is called sacrificial anode. The corroded sacrificial anode block is replaced by a fresh one. Metals commonly employed as sacrificial anodes are magnesium, zinc, aluminium and their alloys. Magnesium has the most negative potential and can provide highest current output and hence is widely used in high resistivity electrolytes like soil.

Applications:

1. Protection as buried pipelines, underground cables from soil corrosion.
2. Protection from marine corrosion of cables, ship hulls, piers etc.
3. Insertion of magnesium sheets into the domestic water boilers to prevent the formation of rust.
4. Calcium metal is employed to minimize engine corrosion.



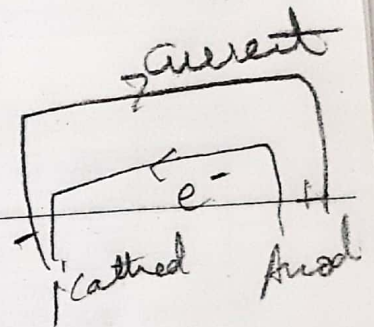
In cathodic protection, an anode of a more strongly reducing metal is sacrificed to maintain the integrity of the protected object (eg., a pipeline, bridge, ship hull or boat).

Advantages:

1. Low installation and operating cost.
2. Capacity to protect complex structures.
3. Applied to wide range of severe corrodents.

Limitations:

1. High starting current is required.
2. Uncoated parts cannot be protected.
3. Limited driving potential, hence, not applicable for large objects.



IMPRESSED CURRENT CATHODIC PROTECTION METHOD

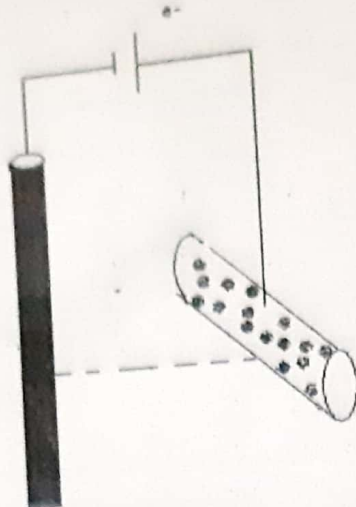
In this method, an impressed current is applied in opposite direction to nullify the corrosion current and convert the corroding metal from anode to cathode. #

Usually the impressed current is derived from a direct current sources (like battery or rectifier on AC line) with an insoluble, inert anode (like graphite, scrap iron, stainless steel, platinum or high silica iron).

A sufficient DC current is applied to an inert anode, buried in the soil (or immersed in the corroding medium) and connected to the metallic structure to be protected. The anode is, usually, a back fill, composed of coke breeze or gypsum, so as to increase the electrical contact with the surrounding soil.

Impressed current cathodic protection has been applied to open water box coolers, water tanks, buried oil or water pipes, condensers, transmission line towers, marine piers, laid up ships etc.

This kind of protection technique is particularly useful for large structures for long term operations.



In Impressed-current cathodic protection, electrons are supplied from an external cell so that the object itself becomes cathodic and is not oxidized.

Comparison of Sacrificial anode method with Impressed current cathodic method:

Sl. No.	Sacrificial Anode method	Impressed Current method
1/	External power supply is not required.	External power supply is required.
2/	The cost of investment is low.	The cost of investment is high.
3/	This requires periodic replacement of sacrificial anode.	Replacement is not required as anodes are stable.
4/	Soil and microbiological corrosion effects are not considered.	Soil and microbiological corrosion effects are taken into account.
5/	This is the most economical method especially when short term protection is required.	This is well suited for large structures and long term operations.
6/	This is a suitable method when the current requirement and the resistivity of the electrolytes are relatively low.	This is a suitable method even when the current requirement and the resistivity of the electrolytes are high.

5

MODIFYING THE ENVIRONMENT-CORROSION CONTROL

Environment plays a major role in the corrosion of metals. Hence, we can prevent corrosion to a great extent by modifying the environment. Some of the methods are

(a) i) Deaeration:

Fresh water contains dissolved oxygen. The presence of increased amount of oxygen is harmful and increases the corrosion rate. Deaeration involves the removal of dissolved oxygen by increase of temperature together with mechanical agitation. It also removes dissolved carbon dioxide in water

ii) By using inhibitors:

Inhibitors are organic or inorganic substances which decrease the rate of corrosion. Usually the inhibitors are added in small quantities to the corrosive medium. Inhibitors are classified into

- 1) Anodic inhibitors (chemical passivators)
- 2) Cathodic inhibitors (adsorption inhibitors)
- 3) Vapour phase inhibitors (volatile corrosion inhibitors)

Anodic Inhibitors:

Inhibitors which retard the corrosion of metals by forming a sparingly soluble compound with a newly produced metal cations. This compound will then adsorb on the corroding metal surface forming a passive film or barrier. Anodic inhibitors are used to repair

- a) the crack of the oxide film over the metal surface
- b) the pitting corrosion
- c) the porous oxide film formed on the metal surface.

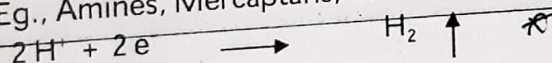
Examples: Chromate, phosphate, tungstate, nitrate, molybdate etc.

Cathodic Inhibitors:

(slows down cathodic reaction)

Depending on the nature of the cathodic reaction in an electrochemical corrosion, cathodic inhibitors are classified into

- a) **In an acidic solution:** the main cathodic reaction is the liberation of hydrogen gas, the corrosion can be controlled by slowing down the diffusion of H^+ ions through the cathode. Eg., Amines, Mercaptans, Thiourea etc.



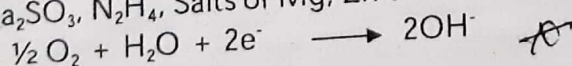
- b) **In a neutral solution:** in a neutral solution, the cathodic reaction is the adsorption of oxygen or formation of hydroxyl ions.

The corrosion is therefore controlled either by eliminating oxygen from the corroding medium or by retarding its diffusion to the cathodic area.

The dissolved oxygen can be eliminated by adding reducing agents like Na_2SO_3 .

The diffusion of oxygen can be controlled by adding inhibitors like Mg, Zn or Ni salts.

Eg., Na_2SO_3 , N_2H_4 , Salts of Mg, Zn or Ni.



Vapour phase inhibitors:

These are organic inhibitors which are readily vapourised and form a protective layer on the metal surface.

These are conveniently used to prevent corrosion in closed spaces, storage containers, packing materials, sophisticated equipments etc.

Examples are Dicyclohexylammonium nitrate, dicyclohexyl ammonium chromate, benzotriazole, phenylthiourea etc.

passive?

9 ANODIC PROTECTION

This is an electrochemical method of corrosion control in which an external potential control system, called potentiostat, is used to produce and maintain a thin non corroding, passive film on a metal or an alloy. The use of potentiostat is to shift corrosion potential into passive potential so that the corrosion of the metal is stopped.

The potential of the object (say acid storage tank) to be protected is controlled by potential controller (potentiostat) so that under certain potential range, the object becomes passive and prevents further corrosion. This potential range depends upon the relationship between the metal and the environment.

Applications:

1. Used in acid coolers in dilute sulphuric acid plants
2. used in storage tanks for sulphuric acid
3. used in chromium in contact with hydrofluoric acid

Limitations:

1. This method cannot be applied in the case of corrosive medium containing aggressive chloride.
2. This cannot be applied if protection breaks down at any point, it is difficult to reestablish.

✓ PROTECTIVE COATINGS

Introduction:

In order to protect metals from corrosion, it is necessary to cover the surface by means of protective coatings. These coatings act as a physical barrier between the coated metal surface and the environment. They afford decorative appeal and impart special properties like hardness, oxidation resistance and thermal insulation.

Classification:

Protective coatings can be broadly classified into two types. They are

1. Inorganic coatings

2) Organic coatings

Inorganic coatings are further classified into two types. They are

i) **Metallic coating:**

1. Hot dipping- Galvanising, Tinning
2. Metal cladding
3. Cementation- Sherardising, Chromising, Calorising
4. Electroplating.

5.9 PROTECTION AGAINST CORROSION (CORROSION CONTROL)

The process of corrosion is slow but the losses incurred are enormous. In fact our economy would be drastically changed if there were no corrosion. But while the corrosion is inevitable, its cost can be considerably reduced. The main responsibility of a corrosion engineer is to control corrosion. Corrosion engineering mainly deals in applying scientific principles to prevent or control corrosion damage.

Diff. Methods for Controlling Corrosion

The various methods by which corrosion can be controlled and prevented are:

1. Protection by proper designing.
2. Material selection.
3. Modifying the environment.
4. Modification of the properties of the metal.
5. Use of inhibitors.
6. Use of protective coatings.
7. Cathodic protection or Electrochemical protection.

These methods are discussed below in detail.

1. Protection by Proper Designing

For controlling the corrosion, proper designing of the equipment plays a significant role. Some of the factors which should be considered for proper designing are:

- (i) The contact between different metals and alloys should not be made, especially if they are far apart in the galvanic series. Where it cannot be avoided delete an efficient electrical insulator should be used. If this is not followed then the more active metal will get corroded causing local corrosion.

Example: Plastic washers are preferred over metal washers because plastic washers act as insulators and prevent corrosion.

- (ii) The rate of corrosion also depends upon the relative sizes of the anode and the cathode. As far as possible, if two dissimilar metals are in contact, the combination of small anode and large cathode should be avoided. (refer to section 5.9) Screws, bolts, nuts and welds should therefore be made from a more noble material than the materials which are joined through them. Thus the screws nuts and bolts will act as cathodes and because of their small areas (cathodes), a limited cathodic reaction occurs leading to decreased rate of corrosion.

Example: Copper rivets are used to fasten steel sheets and not vice versa because copper is nobler than steel and hence acts as cathode. Due to the small area of the copper rivets, the cathodic reaction is limited and anodic reaction of steel proceeds slowly. Whereas if steel rivets are used for joining copper sheet, the small areas of the rivet anode will keep on furnishing electrons to the large cathodic copper sheet thereby corroding steel rivets completely.

- (iii) The presence of crevices between adjacent parts of structure, even in the case of the same metal, should be avoided. The welded butt joints of the same metal/alloy should be used instead of riveted or bolted joints otherwise crevice corrosion may result. In other words welding is better joining technique than mechanical fasteners.

structures are imp

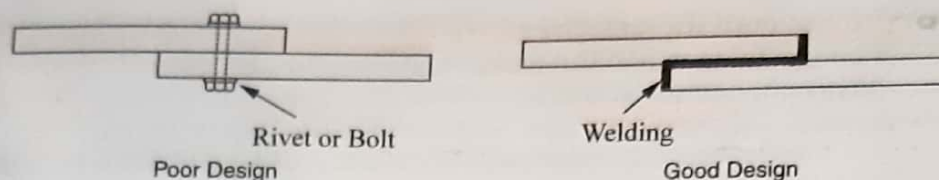


Fig. 5.15 Different techniques of joining the metals/alloys

- (iv) As moisture is the main factor governing corrosion, the design should be such that retention of moisture should be as low as possible. Properties should be so designed that water is drained completely. For that, sharp corners should be avoided.

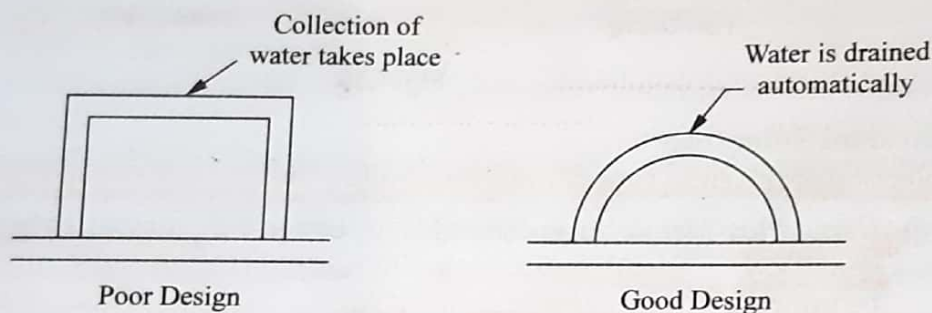


Fig. 5.16 Water stagnation

Storage containers should be designed in such a way that they can be completely drained and cleaned.

- (v) Sharp corners and recesses should be avoided because dust or solid particles may accumulate leading to the corrosion by differential aeration.

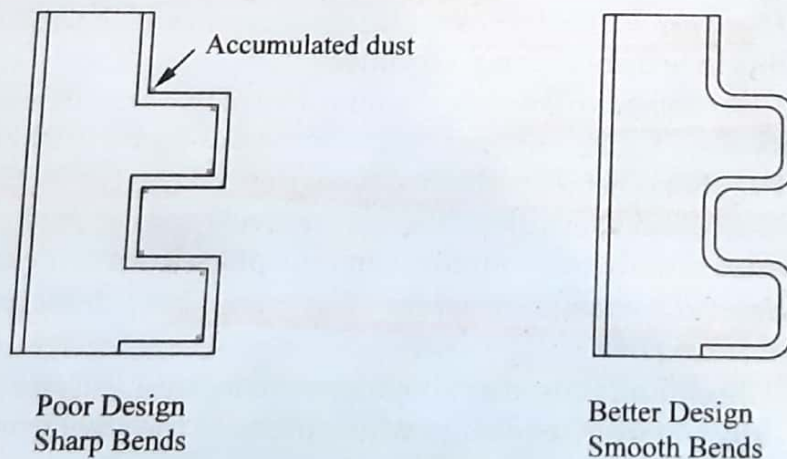


Fig. 5.17

- (vi) The anodic parts should be readily replaceable or should be made thicker for longer life.
- (vii) Increasing pipe diameters helps from mechanical standpoint, by decreasing velocity of the flowing fluids and ensuring laminar flow (refer to section 5.9 for the effects of velocity) Pipelines should be free from obstruction or crevices, in which water may stay giving rise to corrosion by differential aeration.
- (viii) The equipment should be supported on stands to allow circulation of air.

- (ix) Insulation of the materials should be proper otherwise leakage of current takes place. The site from which the leakage takes place becomes anodic and leads to corrosion (stray-current corrosion.)

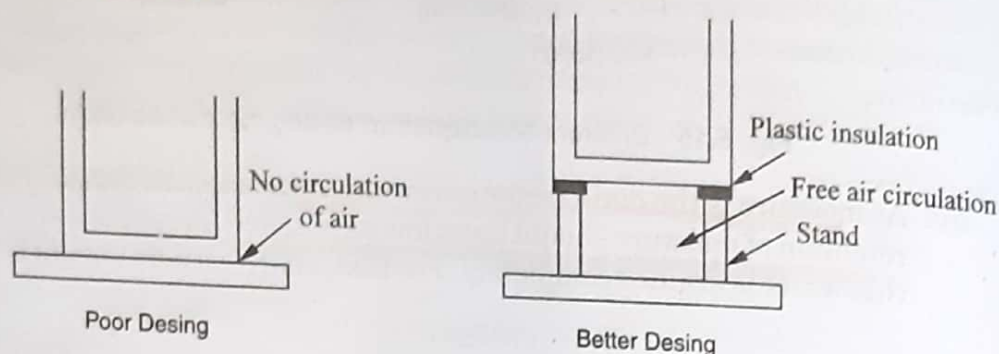


Fig. 5.18

2. Material Selection

Selection of the right type of material is one of the main factors for corrosion control.

- (i) Using Pure Metal: As is discussed in section 5.8, impurities in a metal cause heterogeneity, which lead to corrosion. Thus, the corrosion resistance of a given metal may be improved by increasing its purity.

However it is not always fruitful to use a metal of high purity because:

- Pure metals are expensive
- They are usually weak and do not possess the required strength. For example, one would not build a bridge of pure iron.

- (ii) Using metal alloys: Corrosion resistance of most commercial metals can be increased by alloying them with suitable alloying elements. (Alloys are mixtures of two or more metals or elements.) There are two kinds of alloys:

- Homogeneous alloys are solid solutions in which the components are completely soluble in one another e.g., stainless steel.
- Heterogeneous alloys are the mixtures of two or more separate phases. The components of such alloys are not completely soluble and exist as separate phases.

The choice between the two types depends on the mechanical properties desired. Solid-solution alloys are more corrosion resistant than alloys with two or more phases because no galvanic coupling can take place in former case.

Several corrosion-resistant alloys have been developed depending upon the desired properties.

- Chromium is used as alloying metal for iron and steel as it produces an exceptionally coherent oxide film which protects the steel from further attack.
- Silicon iron alloy called (15% Si and 85% Fe) is highly resistant to the acids because of the formation of silicon oxide film on its surface.
- Cupro-nickel alloys (70% Cu + 30% Ni) containing 0.2% Fe are used for condenser tubes and for bubble trays used in fractionating columns.

3. Use of Inhibitors

Corrosion inhibitors are the substances which when added in small quantities to the corrosive environments decrease the corrosion rate. Inhibitors are organic as well as inorganic substances that dissolve in the corroding medium.

Inhibitors are generally classified on the basis of electrode at which their action is predominant. Thus we have two types of inhibitors, Cathodic inhibitors and Anodic inhibitors.

Alloys

Home

Hetero

definition

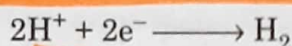
- (i) Anodic Inhibitors: The anodic reaction takes place when the ions from the anodic metal pass into the solution. (For example, $\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$). The anodic inhibitors are the compounds which react with these metal ions to produce insoluble compounds. This insoluble compound gets adsorbed on the metal surface forming a film or a protective barrier thereby reducing the corrosion rate.

The examples of anodic inhibitors include molybdates, phosphates and chromates of alkali and transition elements. Substances like Na_2CrO_4 , NaNO_2 act as anodic inhibitors either by forming less soluble compounds with the anodic metal ions (as discussed above) or they may also retard the corrosion by oxidative repair of the oxide film.

The organic compounds can also act as anodic inhibitors and example include sodium benzoate.

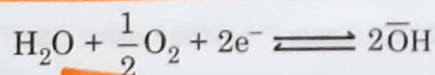
Imp Limitations of anodic inhibitors: Anodic inhibitors are not considered satisfactory due to the following reasons:

- The inhibitors can prevent corrosion only above a certain critical concentration of solution.
 - If insufficient concentration is used then the film so formed may not cover all the anodic area and corrosion of small areas left exposed suffer intense corrosion and cause severe pitting. (due to small anodic area and large cathodic area)
- (ii) Cathodic Inhibitors: The cathodic reaction in acidic medium is represented by:



Thus corrosion can be controlled by slowing down the diffusion of H^+ ions to the cathode. The diffusion of H^+ ions is considerably reduced by organic inhibitors such as amines, mercaptans, heavy metal soaps, substituted ureas and thioureas. These substances are capable of being adsorbed at the metal surfaces and thus restrict the diffusion of H^+ ions. Antimony or arsenic oxides are used as cathodic inhibitors because of the formation of adherent film on the surface of the metal.

In neutral solutions, the cathodic reaction is



In this environment, the corrosion can be controlled by either removing the oxygen from the corroding medium or by not conducting the electrons.

The removal of oxygen is achieved by adding reducing agents such as sodium sulphite or by de-aeration.

ZnSO_4 and MgSO_4 can also be added which react with OH^- ions and the respective hydroxides precipitated on the cathode surface. The important cathodic inhibitor is $\text{Ca}(\text{HCO}_3)_2$ which is found in water.



Imp [Compared to anodic inhibitors cathodic inhibitors are quite safe. This is because, even if they are used in concentrations less than required for complete protection, the partially covered cathodic sites decrease the surface of the cathode whereas anodic area remains practically the same. Thus corrosion decreases due to area effect.

- (iii) Vapour Phase Inhibitors: These are the inhibitors which readily vapourize and form a protective layer of inhibitor on the metal surface. Vapour phase inhibitors (VPI) are used to prevent corrosion in enclosed spaces and also during storage, packing, shipping etc. Small metals are protected from corrosion by keeping them in the envelopes made from paper impregnated with a suitable vapour phase inhibitor.

Dicyclohexylammonium nitrile and cyclohexyl amine carbonate are amongst the most widely used VPIs.

4. Cathodic Protection or Electrochemical Protection

For the structures immersed in soils or electrically-conducting liquids, the cathodic protection is the best way to control corrosion. Cathodic protection is to force the metal to be protected to behave like a cathode as a result of which it is not corroded.

Cathodic protection can be achieved in two ways:

- (i) By using galvanic or sacrificial anode (Galvanic Protection).
- (ii) By using impressed current.

(i) **Galvanic Protection:** In galvanic protection or sacrificial anodic protection, the metal structure to be protected is made the cathode by connecting it by a wire to a more anodic material. As a result, the corrosion takes place at anodic material and the metal structure is saved from corrosion. The more active metal used for the purpose is known as *sacrificial anode*. The sacrificial anode is replaced by a fresh one when it gets corroded or consumed from time to time. Most commonly metals used for sacrificial anodic protection are Mg, Zn, Al and their alloys.

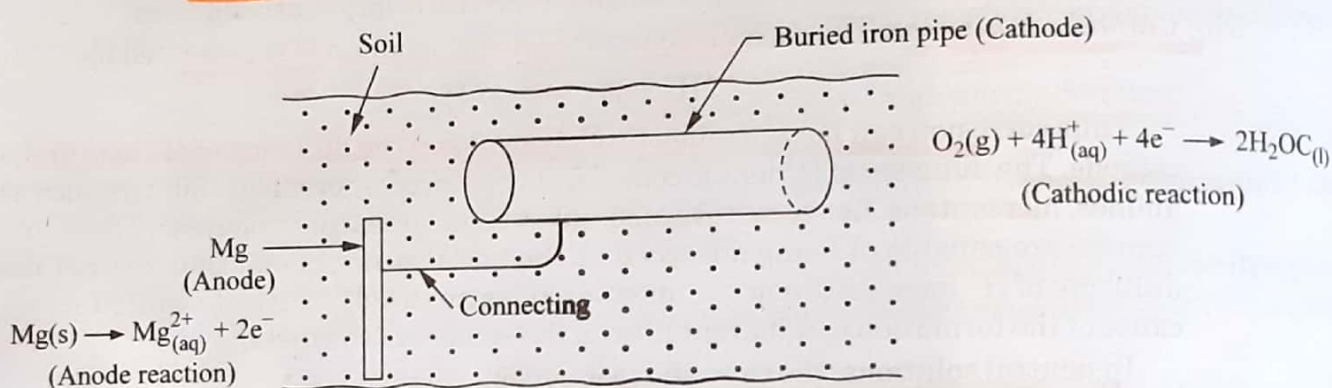


Fig. 5.19 Galvanic Protection using Sacrificial Anode

The application of cathodic protection include:

- (a) Protection of buried pipelines, underground cables.
- (b) Protection of marine structures, ship hulls etc.
- (c) Magnesium rods are inserted into domestic water boilers or tanks to prevent rusting.
- (ii) **Impressed Current Cathodic Protection:** This method, known as electrolytic method of cathodic protection, the object to be protected is made cathode by connecting it to negative terminal of DC source. The positive terminal is connected to a insoluble anode like graphite, scrap iron or platinum. The objective is to apply the current from an external source (insoluble anode) in the opposite direction to that of the corrosion current. The insoluble anode is buried in the soil or in the conducting medium adjacent to the metal which is to be protected (Fig. 5.20) These are placed in such a manner that a uniform potential is obtained over the entire metal surface. The electrons flow to the metal structure and as a result it act as cathode and is protected. The anode has to be replaced periodically.

The impressed current cathodic protection has been applied to protect.

- (a) Buried pipelines, transmission line towers.
- (b) Water pipes, water tanks, marine piers etc.

This kind of technique is particularly useful for large structures which require long term applications.

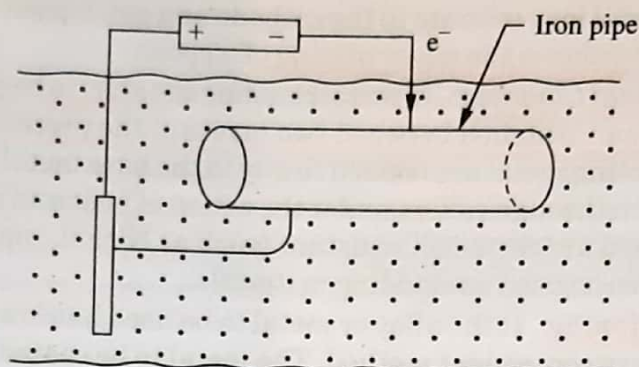


Fig. 5.20 Impressed Current Cathodic Protection

5. Anodic Protection

Anodic protection is applicable to the metals which can be passivated by the deposition of oxide films. In anodic protection, a protective oxide film is grown on the surface of the metal structure to be protected. This is done by electrochemical means i.e., by applying current in a direction that makes the metal structure more anodic. Under these conditions the metal structure is passivated when the required potential is maintained and hence rate of corrosion decreases.

Anodic protection has been applied in case of steel, stainless steel aluminium and chromium. It must be remembered that potential has to be accurately maintained so that it just induces passivity. 2ex.

6. Protective Coatings

The metal structure to be protected can be coated with metallic or non-metallic coatings. These coatings insulate the insulate from the surrounding corrosive environment. The various types of coatings used for controlling corrosion are:

(i) Metallic Coatings

(ii) Non-metallic Coatings

(i) Metallic Coatings: The metals used for coatings are Zn, Ni, Sn, Cd and Cr because these are capable of forming their respective oxide films which adhere to the surface of the metal and prevents corrosion. For example, to prevent the rusting of iron it can be coated with Zn or Cr.

Metallic coatings can be applied by using following techniques

(a) Electroplating: This is the most widely used method of coating metals to protect them from corrosion. The metal to be plated is made the cathode in a plating bath

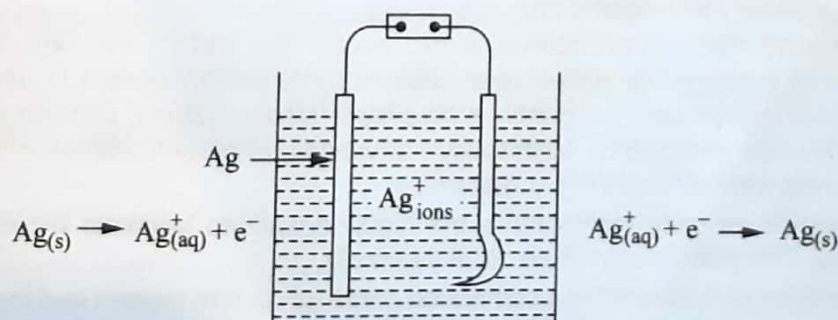


Fig. 5.21 Electroplating

containing the ions of the metal to be deposited. The anode is either an inert material or coating metal to be deposited. When current is passed, the coating metal ions migrate to the cathode and get deposited in the form of thin layer. Fig. 5.16 shows the silver plating of a spoon

- Imp* ✓
- (b) **Metal Cladding:** In metal cladding the alloy to be protected, (known as base metal) is sandwiched between two layers of the protecting metal. These layers of protecting metal are bonded firmly to the base metal by pressing the whole combination through rollers under the action of heat and pressure. The cladding material used are corrosion resistant (such as Nickel, copper, lead, silver etc.) Alloys can also be used as cladding materials.
 - (c) **Dipping:** If the alloy or metal to be used has low melting point then dipping is a very convenient method. The metal to be coated is passed through a bath of the molten coating metal and the thickness of the coating is adjusted by squeezing out the excess of the coating metal with rollers. Coating of zinc on iron by this process is called galvanizing.
 - (d) **Spraying:** A stream of 'atomized' droplets is blown on to the surface of the metal to be protected. This type of coating can be applied to the finished structures such as tanks, ships and towers etc.
 - (ii) **Non-Metallic Coatings:** The non-metallic coatings may be inorganic or organic coatings and fall under the following categories.
 - (a) **Organic Coatings:** Organic coatings are inert organic barriers which provide corrosion resistance as well as decorative value to the structure which is coated. The organic coatings include; paints, varnishes, enamels, and lacquers.
 - (b) **Chemical Conversion Coatings:** These are inorganic surface-barriers produced by chemical or electrochemical reactions at the surface of the metal. These coatings provide the protection to the base metal from corrosion and also provide a surface ideal for painting or other decorative measures. Examples of chemical coatings include, phosphate coating, chromate coating etc. The phosphate coating is produced by the chemical reaction of base metal with aqueous solution of phosphoric acid and zinc or manganese phosphate. Such coatings are applied to iron, steel and zinc. The chemical reaction between the phosphating solution and the base metal results in the formation of a surface film consisting of crystalline zinc-iron or manganese-iron phosphates.

Case Study **ASHOKA PILLAR**

This pillar was built by samrat Ashoka who ruled around 280 B.C, and is known by his name Ashoka Pillar. It is situated near Qutab Minar in Delhi. This pillar is the testimony of the advanced metallurgical technology ancient India possessed.

The pillar is solid shaft of iron, sixteen inches in diameter and 23 feet high. What is most astounding about it is that it has never rusted even though it has been exposed to wind and electrolysis! The technique used to cast such a gigantic, solid pillar is also a mystery, as it would be difficult to construct another of this size even today. With today's advances only four foundries in the world could make this piece and none were able to keep it rust free.

The pillar stands as mute testimony to the highly advanced scientific knowledge that was known in ancient India. The pillar is said to be 1600 years old.

A protective layer of 'misawite'—a compound made up of iron, oxygen and hydrogen on the steel pillar, which is said to contain phosphorus- is claimed as the reason for the non-corrosive existence.